

David Cropper

Melbourne, VIC | Australian Citizen

dave_cropper2003@yahoo.co.uk | 0400 811 545

LinkedIn: <https://www.linkedin.com/in/david-cropper/>

Portfolio: <https://journey.chaseriskandcompliance.com.au/>

GitHub: <https://github.com/dcrops>

Professional Summary

AI Engineer with **3.5 years of commercial enterprise Data Engineering experience** and a portfolio of production-minded AI systems spanning Retrieval-Augmented Generation (RAG), operational intelligence, explainable AI and enterprise decision-support.

Experienced designing and building AI applications using **Python, FastAPI, Docker, OpenAI APIs and modern AI engineering practices**, combining deterministic software engineering with Large Language Models to produce transparent, evidence-backed outcomes.

I approach AI engineering as a software engineering discipline, emphasising architecture, testing, explainability and maintainability through disciplined AI-assisted development practices.

Interested in engineering AI systems that deliver measurable operational outcomes within collaborative product and engineering teams.

Selected Engineering Highlights

- Designed and built **four production-minded AI applications** spanning Retrieval-Augmented Generation, operational intelligence, explainable AI and enterprise decision-support.
- Developed Python-based AI systems using **FastAPI, Docker, OpenAI APIs, LangChain and vector databases**.
- Built AI systems that combine deterministic analytics with Large Language Models to generate **transparent, evidence-backed recommendations**.
- Applied architecture-first AI engineering practices using **Cursor, Git, PyTest and milestone-driven development**.
- Created a public AI engineering portfolio demonstrating architecture, implementation, testing and engineering decision-making

Professional Experience

Chase Risk & Compliance

AI Engineer – Independent Research & Development

Independently designed and developed a portfolio of production-minded AI systems to deliberately transition from enterprise Data Engineering into AI Engineering, applying modern software engineering principles throughout the complete development lifecycle.

Engineering Responsibilities

- Designed and developed production-minded AI applications using Python, FastAPI, OpenAI APIs and Docker.
- Applied architecture-first engineering practices using Cursor, Git and PyTest.
- Built systems focused on explainability, operational intelligence and enterprise decision support.
- Documented architecture, testing strategies and engineering decisions through a public portfolio.

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Data Engineer | Mar 2020 – Oct 2023

- Developed enterprise data pipelines and operational reporting solutions.
- Performed SQL transformations and data validation.
- Worked in Agile teams supporting production systems.
- Used Python, SQL, Apache NiFi, Git and PyTest.

Featured AI Projects

Operational Intelligence Copilot

Purpose: Designed a domain-agnostic operational intelligence platform that combines deterministic analytics with Large Language Model reasoning to analyse structured business data and generate explainable, evidence-backed executive insights. The platform was built to explore how AI can augment operational decision-making while maintaining transparency and evidence traceability.

Engineering Highlights

- Intelligent intent routing
- Dataset profiling
- Evidence-backed executive reporting
- Explainable AI recommendations

- Anomaly detection
- Regression-tested analytics workflows

Technology Stack: Python • FastAPI • OpenAI APIs • PyTest • Git

Governance-Aware Document Intelligence RAG

Purpose: Enterprise Retrieval-Augmented Generation platform supporting trustworthy AI responses over organisational knowledge.

Engineering Highlights

- Multi-format document ingestion
- Embedding generation
- Retrieval orchestration
- Evaluation framework
- Telemetry
- Explainability
- Grounding validation

Technology Stack: Python • FastAPI • Docker • LangChain • ChromaDB • OpenAI • PyTest • Git

Payroll Diagnostics Engine

Purpose: Rule-based operational decision-support platform designed to identify payroll anomalies through deterministic business rule evaluation, producing explainable operational findings and executive-ready reporting.

Engineering Highlights

- Rule engine
- Explainable diagnostics
- Executive reporting
- Operational analytics
- Automated validation

Technology Stack: Python • Pandas • OpenPyXL • PyTest • Git

Public Holiday Entitlements Application

Purpose: Business rules platform combining geospatial processing and external APIs to determine Australian public holiday entitlements.

Engineering Highlights

- Geospatial logic
- Business rules
- API integration
- Compliance reporting

Technology Stack: Python • REST APIs • Geospatial Processing • Git

AI-Assisted Engineering Practices

I apply AI to **improve engineering quality**, not replace engineering judgement.

Planning

Architecture-first design • Milestone planning • Acceptance criteria • Cursor Plan Mode

Implementation

Cursor Agent Mode • Iterative feature development • Controlled implementation boundaries

Quality

PyTest regression testing • Behavioural testing • Evidence-based debugging

Review

Cursor Ask Mode • Architecture review • Maintainability review • Refactoring

Delivery

Git version control • Small, reviewable commits • Documentation-first development

Certifications & Professional Development

- AWS Certified Developer – Associate
- LangChain & LLM Application Development
- Microsoft Fabric, Databricks, Snowflake, dbt (professional development)
- Azure Data Factory (professional development)